FPGA-BASED REAL-TIME OSCILLOSCOPE

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DIGITAL SYSTEM DESIGN

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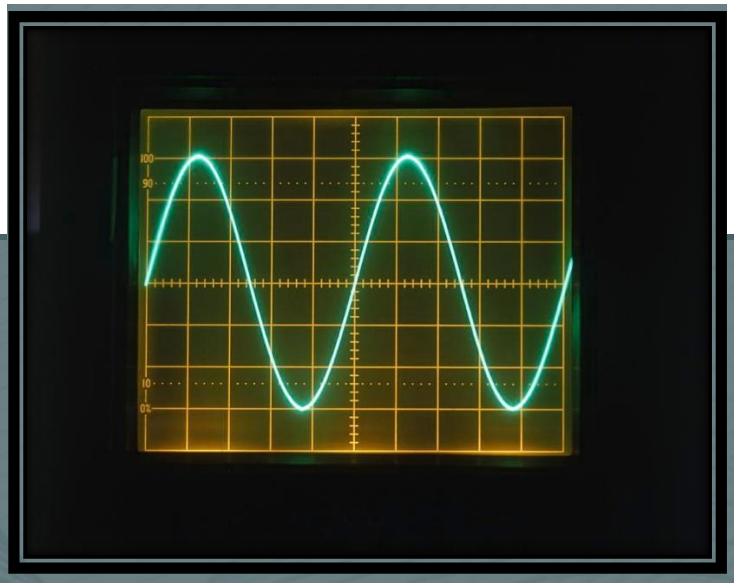
INTRODUCTION

We have chosen this project because we want to deepen your understanding of digital systems.

Oscilloscopes combine ADCs, buffering, triggering, and display logic. It is one of the most fundamental tools in electronics, and building one from scratch provides a deeper understanding of how measurement instruments work internally. Instead of treating it as a black box, this project lets us reverse-engineer its essential functions and re-implement them using the DEO-Nano-SoC FPGA.



Objective

The objective of this project is to design and demonstrate a functional, real-time digital oscilloscope implemented primarily in the FPGA fabric of the DE0-Nano-SoC.

At the final demo, the system will acquire an analog input signal through the TI Educational Booster Pack's onboard ADC, sample it continuously, and display the resulting waveform on an external monitor using a custom VGA controller. The VGA output will be generated using a custom RTL VGA timing controller. Although Intel provides a VGA IP core, it is designed for full video pipelines and framebuffer-based systems and would not provide the low-level timing control or RTL design experience required for this project.

The oscilloscope will support basic features expected from a low-frequency measurement instrument, including adjustable trigger level, selectable trigger mode, and a configurable time base that controls how many samples are shown on the screen.

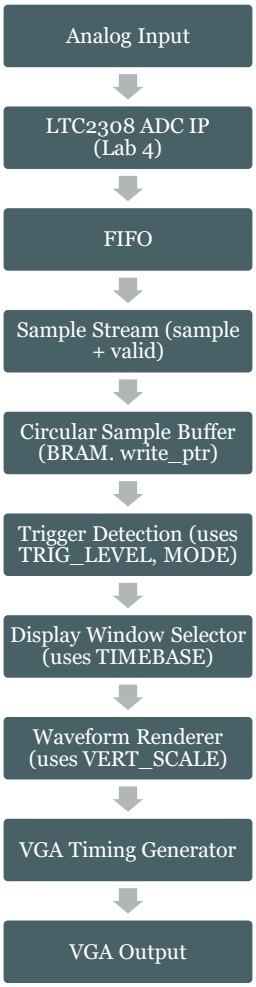
The demonstration will show the oscilloscope capturing and displaying common low-frequency waveforms such as sine, square, and triangle waves. It will also be capable of displaying the output of simple analog circuits such as op-amp filters, provided the input is conditioned to remain within the ADC's 0–3.3 V range. The signal-processing path including sampling, buffering, triggering, and waveform rendering will be implemented entirely in hardware using System Verilog. The ARM processor may be used for optional user-interface control, but it will not participate in the real-time data path.

By the end of the project, the system will demonstrate a complete hardware pipeline from analog input to digital display, showing that the FPGA can perform real-time acquisition, processing, and visualization of signals without relying on software intervention.

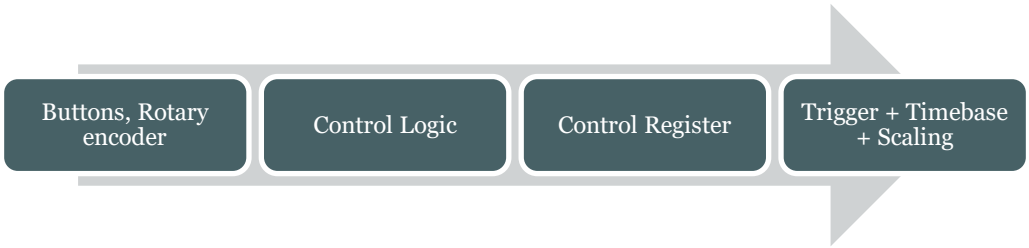
Detailed Description (Processes and Block Diagrams)

Oscilloscope Architecture

Data Path



Control Path



Control Register Map

Register	Width	Purpose
TRIG_LEVEL	12 bits (same as ADC bus)	Sets the ADC threshold
TRIG_MODE	2 bits	Rising, Falling, Level trigger
TIMEBASE	4-8 bits	Samples per screen (256, 512, 1024,..4096)
VERT_SCALE	3-4 bits	Vertical zoom (extended feature)
CONTROL	4 bits	Enable/reset/force trigger

Schedule & Milestones

Week 1

- Integrating Lab 4 into the project.
- Configuring ADC to sample a single channel continuously
- Reading samples from FIFO and writing into circular buffer.
- Verifying buffer write pointer and wrap-around behavior in simulation.

Week 2

- Implementing rising-edge trigger detection.
- Adding control registers
- Simulating trigger behavior.

Week 3

- Implementing 680*480 VGA timing generator.
- Display test patterns (color bars, grid, etc.).
- Generates HSYNC, VSYNC, and pixel coordinates for a standard mode.

Week 4

- Keep working on waveform renderer (sample to pixel mapping).
- Integrate circular buffer -> renderer -> VGA.
- Implement TIMEBASE and VERT_SCALE registers.
- Display live waveform on monitor.

Week 5

- Full system integration.
- Prepare final demonstration.

Risk Management

External ADC not used

Using an external high-speed ADC was intentionally avoided. External ADCs introduce significant mixed-signal risks, including analog front-end design, reference voltage stability, high-speed SPI timing, and PCB-level signal integrity. These issues fall outside the scope of the course and would jeopardize the project timeline. Therefore, we will be displaying low frequency waveforms.

But we will still order the external ADC chip just in case.

ADC Timing or FIFO Underflow

Risk:
ADC sampling may not align with buffer read timing, causing underflow.

Mitigation:

- Use Lab 4 FIFO to decouple ADC and oscilloscope pipeline
- Monitor underflow bit for debugging
- Increase buffer size if needed

Trigger Logic Instability

Risk:
Trigger may fire too often or not at all.

Mitigation:

- Add hysteresis or simple debounce
- Provide manual “force trigger” bit
- Allow user to freeze display for debugging

Integration Complexity

Risk:
Modules may work individually but fail when combined.

Mitigation:

- Follow bottom-up integration
- Test each module with testbenches
- Add debug LEDs or UART prints if needed

References

1. *FPGA-based oscilloscope*. (2007). Cornell University ECE 5760 Final Projects.FPGA Digital Oscilloscope

2. D. Wu, H. Jiao, Y. He and B. Lin, "Innovative Experimental Project Design of a Digital Oscilloscope Using FPGA Technology," 2024 IEEE 2nd International Conference on Electrical, Automation and Computer Engineering (ICEACE), Changchun, China, 2024, pp. 312-316, doi: 10.1109/ICEACE63551.2024.10898918. keywords: {Costs;Oscilloscopes;Electric variables measurement;User interfaces;Software;Hardware;Liquid crystal displays;Frequency measurement;Hardware design languages;Field programmable gate arrays;FPGA;Digital Oscilloscope;Nios II;LCD;Waveform storage},